

Sinkhorn Divergence

$$S_{\varepsilon}(\alpha, \beta) = \text{OT}_{\varepsilon}(\alpha, \beta) - \frac{1}{2}\text{OT}_{\varepsilon}(\alpha, \alpha) - \frac{1}{2}\text{OT}_{\varepsilon}(\beta, \beta)$$

Maximum Mean Discrepancy

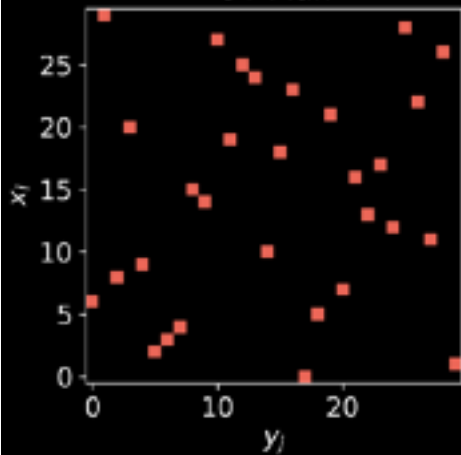
$$\text{MMD}(\alpha, \beta) = \|\mathbb{E}_{x \sim \alpha}[k(x, \cdot)] - \mathbb{E}_{y \sim \beta}[k(y, \cdot)]\|_{\mathcal{H}_k}$$

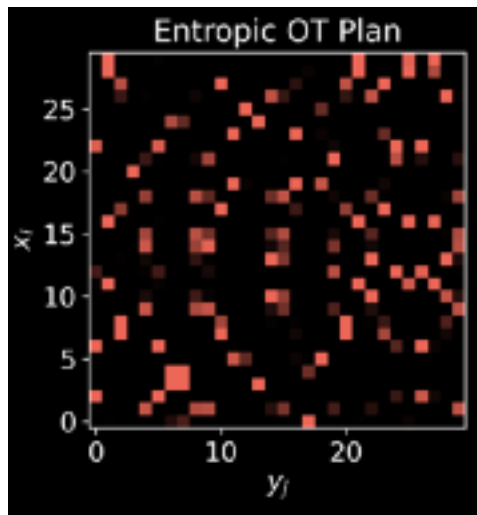
Classic OT

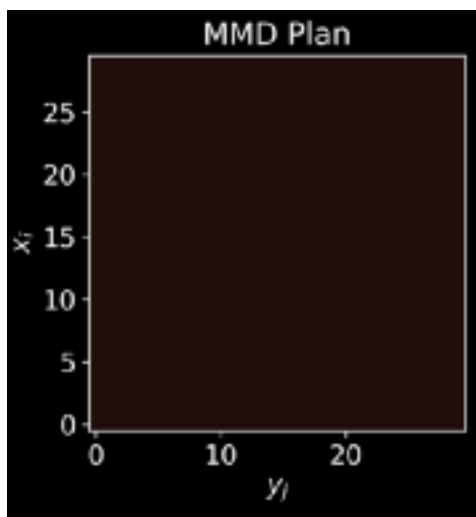
$$\text{OT}(\alpha, \beta) = \min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) \, \mathrm{d}\pi(x, y)$$



OT Plan











Sinkhorn Divergence

$$S_{\varepsilon}(\alpha, \beta) = \text{OT}_{\varepsilon}(\alpha, \beta) - \frac{1}{2}\text{OT}_{\varepsilon}(\alpha, \alpha) - \frac{1}{2}\text{OT}_{\varepsilon}(\beta, \beta)$$